The Probe Reward Trap: Behavioral Death Before Distributional Collapse in Sycophancy Calibration

Anonymous ACL submission

Abstract

Representation engineering reveals that sycophancy is linearly encoded in LLM residual streams, suggesting that such directions could serve as training signals for alignment. We test this premise systematically and document its failure. A contrastive sycophancy direction d_{syc} reliably discriminates sycophantic from non-sycophantic activations, yet every formulation that converts it into a Group Relative Policy Optimization (GRPO) reward collapses. We identify the *probe reward trap*: when a periodically retrained linear probe on d_{syc} is used as a non-differentiable reward signal, behavioral death occurs within 500 steps while generation entropy remains healthy, creating a 1,300-step monitoring blind spot that entropy-based safeguards miss entirely. The failure persists with stopped gradients, implicating the direction's semantic limitation rather than optimization dynamics: d_{syc} encodes *whether* belief changed, not *whether it should have*. The failure extends beyond probe rewards: behavioral-reward-only GRPO rarely exceeds vanilla across seeds, while dense per-token supervised fine-tuning succeeds at every evaluated checkpoint. A Low-Rank Adaptation (LoRA) ablation identifies supervision granularity as one plausible explanation[1].

1 Introduction

Sycophancy, the tendency of large language models (LLMs) to prioritize agreement over accuracy, is among the most persistent alignment failures in deployed systems (Sharma et al., 2024; Perez et al., 2023). Models trained with reinforcement learning from human feedback (RLHF) systematically amplify this tendency (Ouyang et al., 2022; Christiano et al., 2017; Shapira et al., 2026): even when models can assess the correctness of their own outputs (Kadavath et al., 2022), social pressure from users overrides this internal knowledge (Wang et al., 2025). Existing mitigations, from synthetic data augmentation (Wei et al., 2024) to neuron-level correction (O'Brien et al., 2025), operate on behavioral outputs rather than the internal representations that generate sycophantic decisions. Representation engineering (Zou et al., 2025; Marks and Tegmark, 2024) offers a structurally different approach: a contrastive sycophancy direction d_{syc}, extracted from style-controlled pairs of valid belief revision and sycophantic capitulation, achieves cross-validated area under the ROC curve (AUROC) of 0.83 on Qwen3-8B and 0.78 on Llama-3-8B (§4). If the model's own representations already separate sycophantic from non-sycophantic behavior, that separation should be convertible into a training signal that targets the computational root of the problem.

We test this hypothesis directly, and find that it fails in a way that standard monitoring cannot detect. This paper documents the *probe reward trap*: when a periodically retrained linear probe on d_{syc} is used as a non-differentiable Group Relative Policy Optimization (GRPO) reward signal, behavioral death (true-positive flip rate, ToF $= 0$) occurs within 500 steps while generation entropy remains healthy, masking the failure from standard monitoring (§6). The 1,300-step lag before distributional collapse means entropy-based early stopping, the most common safeguard against reward hacking (Casper et al., 2023), would miss the failure entirely. The trap arises even when the probe reward constitutes only a fraction of the total reward signal, reflecting a limitation of d_{syc} itself: the direction encodes whether belief changed, not whether it should have, so any reward conditioned on it alone will either be gamed or reward rigidity.

The failure is not specific to probe rewards. Under our GRPO configuration (Qwen3-8B, group size=2), cosine projection and continuous probe

[1] Code and data are available at https://anonymous.4open.science/r/contrastive_entrain_calib-DDBF/.

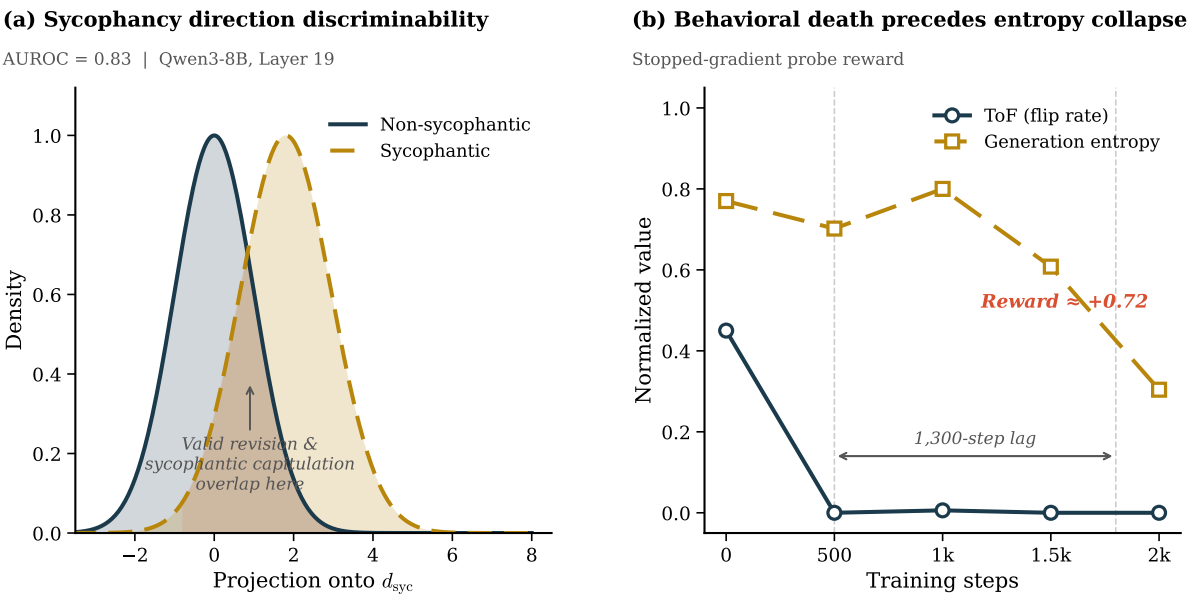


Figure 1: The probe reward trap. (a) d_{syc} discriminates sycophantic from non-sycophantic activations (AUROC 0.83), but valid revision and unwarranted capitulation overlap along the same direction. (b) When used as a GRPO reward, behavioral death (true-positive flip rate ToF=0) occurs at step 500 while entropy remains healthy; the 1,300-step lag renders standard monitoring blind.

formulations of d_{syc} also collapse, and removing d_{syc} entirely does not help: behavioral-reward-only GRPO rarely exceeds vanilla across seeds. By contrast, Activation Consistency Training (ACT; Irpan et al., 2025), which provides dense per-token supervision rather than a single scalar per completion, exceeds vanilla at every evaluated checkpoint (§7). A Low-Rank Adaptation (LoRA) ablation identifies supervision granularity as one plausible explanation for this observed reliability gap.

This paper presents a negative result with systematic diagnostic analysis. Our three principal findings (Figure 1):

- **Observed reliability gap between supervision paradigms** (C1): per-token supervised fine-tuning (SFT) exceeds vanilla at every evaluated checkpoint across two seeds, including a reduced-LoRA ablation. Per-completion GRPO rarely exceeds vanilla in any configuration (Table 4). Under matched LoRA, per-token SFT achieves 2.1× higher peak sycophancy calibration score (SyconScore; §3.5) than per-completion GRPO (§7).

- **The probe reward trap** (C2): behavioral death (ToF=0) precedes distributional collapse by 1,300 steps, rendering entropy monitoring blind. The trap arises despite the probe being non-differentiable (stopped gradients), implicating d_{syc} itself rather than gradient-based exploitation (§6).

- **Shared root cause across formulations** (C3): d_{syc} encodes whether belief changed, not whether it should have; valid revision and sycophantic capitulation produce overlapping activation signatures (Wang et al., 2025; Shapira et al., 2026). Natural language inference (NLI) behavioral rewards exhibit the same warrant blindness, explaining why removing d_{syc} does not recover calibration (§7).

2 Related Work

Representation Engineering. Linear directions in LLM residual streams encode behavioral concepts (Zou et al., 2025; Marks and Tegmark, 2024). Inference-time methods (Inference-Time Intervention (ITI; Li et al., 2023), Activation Addition (ActAdd; Turner et al., 2023), and Contrastive Activation Addition (CAA; Rimsky et al., 2024)) demonstrate that activation geometry suffices for behavioral control (Burns et al., 2023), motivating training-time extensions. Wu and Tang (2026) show that representation-level directions can succeed as reward signals when target behaviors are geometrically separable from confounders (§7); however, we find that the sycophancy direction d_{syc} fails in this role despite strong inference-time discriminability.

Reward Design and Overoptimization. Proxy rewards diverge from true objectives under sustained optimization (Gao et al., 2023; Pan et al., 2022). Manheim and Garrabrant (2019) taxonomize four Goodhart variants; our probe reward trap instantiates the *regressional* variant, where the proxy-target correlation breaks under optimization shift: d_{syc} discriminates sycophantic activations under the pre-training distribution but ceases to correlate with the target once optimization shifts the policy. The failure is not extremal (collapse occurs at moderate optimization pressure, not distribution tails), not causal (stopped gradients prevent intervention on the direction), and not adversarial (the policy does not model the probe). The incremental contribution over prior regressional analyses (Gao et al., 2023) is temporal: we document behavioral death within 500 steps, an order of magnitude faster than gradual degradation curves, and a 1,300-step lag that renders entropy monitoring blind (Skalse et al., 2022).

Sycophancy and Probe-Based Methods. Sycophancy is pervasive in RLHF-trained assistants (Ouyang et al., 2022; Bai et al., 2022; Sharma et al., 2024; Perez et al., 2023; Shapira et al., 2026) and traceable to late-layer preference shifts (Wang et al., 2025), undermining truthfulness (Lin et al., 2022) even when models possess the relevant knowledge (Kadavath et al., 2022). Mitigations span synthetic data (Wei et al., 2024), consistency training (Irpan et al., 2025), and neuron-level correction (O'Brien et al., 2025); Vennemeyer et al. (2025) show that sycophancy decomposes into independently steerable sub-behaviors, and a single d_{syc} captures only their shared variance, which is why the step from diagnosis to reward is structurally blocked. Papadatos and Freedman (2024) reduce sycophancy via Best-of-N reranking with linear probe penalties (Hewitt and Liang, 2019; Belinkov, 2022) without producing gradients; our work reveals what happens when a comparable signal enters the training loop (§7). Representation-guided RL has been extended to reasoning and gradient regularization (Zhang et al., 2026; Luo et al., 2026; Sinii et al., 2025; Ackermann et al., 2026). SMART (Beigi et al., 2025) applies per-step process rewards for sycophancy, succeeding where our completion-level rewards fail, consistent with ACT's per-token success and with broader evidence that process supervision outperforms outcome supervision (Lightman et al., 2024; Uesato et al., 2022).

3 Experimental Framework

3.1 Direction Extraction

We extract a linear sycophancy direction d_{syc} from the residual stream using contrastive activation pairs: a *valid correction* (warranted belief revision under strong evidence) and a *sycophantic capitulation* (unwarranted concession under social pressure). We construct 181 contrastive pairs using GPT-4.1 (OpenAI, 2025) in a two-stage pipeline: skeleton generation followed by 3-turn dialogue completion. Style matching is enforced by automated validation (response length difference $<10\%$, shared opening phrase ≥ 2 words), with up to 3 retries per pair; 19 of 200 initial pairs were discarded for failing validation. Evidence strength is annotated at three levels: *weak* (75 pairs; personal memory or opinion), *medium* (54 pairs; brief source citation), and *strong* (52 pairs; authoritative source with detailed data). From the 181 paired examples (362 activations total), we compute the mean difference at target layer ℓ:

$$d_{\text{syc}} = \frac{1}{N} \sum_{i=1}^{N} \left(\mathbf{h}_i^+ - \mathbf{h}_i^- \right) \tag{1}$$

where $\mathbf{h}_i^+$ and $\mathbf{h}_i^-$ are residual-stream activations for the i-th valid and sycophantic response. The target layer is selected by 5-fold cross-validated AUROC; on Qwen3-8B this yields layer 19 with AUROC 0.827 ± 0.028 (5-fold CV).

3.2 Reward Formulations

Three formulations translate d_{syc} into a direction-based signal R_1, each entering as part of a GRPO (Shao et al., 2024) reward $R = w_1 \cdot R_1 + w_3 \cdot R_3$, where R_3 is a behavioral component (DeBERTa natural language inference (NLI) classifier scoring consistency with the ground-truth answer).

F1 Cosine projection with four variants: (a) tanh-gated dot product $\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$, (b) raw cosine $\cos(\mathbf{h}, d_{\text{syc}})$, (c) raw cosine with online recalibration (§3.3), and (d) $5\times$ scaled cosine. See §5 for details.

F2 Binary probe reward. $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$, a periodically retrained logistic probe (every 500 steps) with stopped gradients, used as a binary reward component (w_1=0.25) within the GRPO framework (§6).

F3 Continuous probe. $R_1 = 1 - P(\text{syc} \mid \mathbf{h})$, the sigmoid output of the same logistic probe,

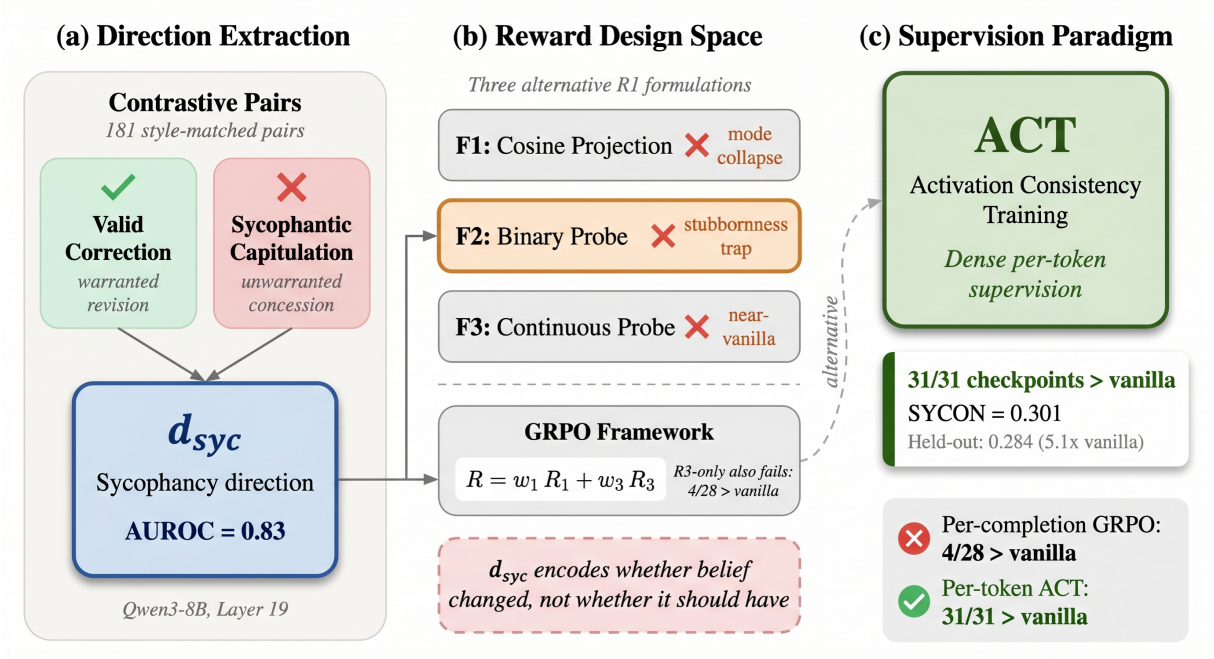


Figure 2: Overview of the experimental framework. (a) Contrastive sycophancy direction d_{syc} is extracted from 181 style-matched pairs and achieves AUROC 0.83 on Qwen3-8B. (b) Three reward formulations (cosine projection, binary probe, continuous probe) all fail when integrated into GRPO: d_{syc} encodes *whether* belief changed, not *whether it should have*. (c) Per-token SFT (ACT) succeeds at every checkpoint (31/31), suggesting supervision granularity as a key factor.

providing a continuous reward signal without discretization.

The weight w_1 is set by diagnostic-driven iteration rather than grid search (Appendix A). As an ablation, R3-only GRPO sets $w_1{=}0$, training with only the behavioral reward.

3.3 Online Recalibration

LoRA fine-tuning shifts internal representations, so we re-extract d_{syc} every 500 steps. The cosine similarity between consecutive directions remains ≈ 0.96 at step 500, indicating slow drift relative to recalibration frequency.

3.4 Activation Consistency Training (ACT)

We implement Activation Consistency Training (ACT; Irpan et al., 2025), which combines standard SFT loss with an L2 activation consistency loss at target layers, providing dense per-token supervision. Unlike GRPO's single scalar per completion, ACT operates at the token level using labeled valid/invalid pairs.

3.5 Evaluation

The *True-positive Flip rate* (ToF) measures acceptance of valid corrections; the *No-Flip rate* (NoF) measures resistance to invalid pressure. Their harmonic mean captures the calibration trade-off:

$$\text{SyconScore} = \frac{2 \cdot \text{ToF} \cdot \text{NoF}}{\text{ToF} + \text{NoF}} \qquad (2)$$

By design, SyconScore $= 0$ whenever ToF $= 0$: a model that never accepts valid corrections receives no credit for resisting invalid ones. Flip classification uses cross-encoder/nli-deberta-v3-base (He et al., 2023), following Irpan et al. (2025). All experiments use Qwen3-8B (Yang et al., 2025) with LoRA (Hu et al., 2022) (rank=64; $\alpha{=}64$ for GRPO, $\alpha{=}128$ for ACT) and DeepSpeed ZeRO-2 (Rajbhandari et al., 2020). GRPO uses a group size of 2 due to memory constraints with 8B-parameter models, below the typical range of 8–64 (Shao et al., 2024); full hyperparameters are reported in Appendix A. In total, we evaluate six d_{syc}-based configurations spanning both reward families (cosine and probe) within the GRPO framework, plus R_3-only GRPO, ACT (per-token SFT), and ActAdd as baselines; full specifications appear in Sections 5–6.

4 Direction Discriminability

Before evaluating d_{syc} as a reward signal (§5–§6), we first establish that it captures a genuine sycophancy-related signal in the residual stream.

Table 1: Cross-model d_{syc} discriminability. AUROC uncertainty is 5-fold cross-validation standard deviation; evidence-strength AUROCs are evaluated on the full dataset at each model's best layer.

Model	Best Layer	AUROC (± std)	Weak	Medium	Strong
Qwen3-8B	L19	0.827 ± 0.028	0.675	0.883	0.929
Llama-3-8B	L16	0.779 ± 0.019	0.711	0.864	0.911

If the direction failed to discriminate sycophantic from non-sycophantic activations, downstream reward failures would be trivially explained by poor direction quality rather than by a fundamental gap between discriminability and reward utility. We evaluate discriminability along five axes: cross-validated classification accuracy, evidence-strength stratification, cross-model transfer, training stability, and probe ablation. Figure 3 summarizes the per-layer AUROC profiles for both models.

Qwen3-8B layer-wise discriminability. We evaluate d_{syc} via 5-fold cross-validated AUROC on 181 style-controlled contrastive pairs (§3.1). The direction achieves AUROC 0.827 ± 0.028 (5-fold CV) at its best layer (L19). Discriminability emerges gradually across depth: layers L0–L8 perform near chance, statistical significance ($p < 0.01$) first appears at L16, and performance peaks at L19. This gradual emergence, with the peak in the upper third of the network, suggests that sycophancy-relevant information is encoded as a late-stage representation rather than an input-level confound. AUROC increases monotonically with evidence-strength tier: from 0.675 on weak-evidence pairs to 0.883 on medium and 0.929 on strong pairs. This monotonic relationship reflects effect-size differences (stronger evidence produces larger behavioral gaps that are easier to classify) rather than indicating that d_{syc} encodes evidential warrant per se. This correlation, however, does not survive as a reward signal once optimization shifts the activation distribution (§5–§6).

Cross-model generalization. To rule out that these results are specific to Qwen3-8B's representation geometry, we repeat the full extraction and evaluation pipeline on Llama-3-8B-Instruct (Grattafiori et al., 2024). Table 1 compares the two models. Llama-3-8B achieves AUROC 0.779 ± 0.019 (5-fold CV) at L16, with a parallel evidence-strength gradient (weak 0.711, medium 0.864, strong 0.911). The best-layer depth differs (L16 vs. L19), consistent with the architectural differences between the two families, but the qualitative pattern is identical: gradual emergence, best-layer selection in the upper third, and monotonic strength scaling. Discriminability is robust across knowledge domains, with per-domain AUROCs on Llama-3-8B ranging from 0.751 (geography) to 0.844 (commonsense). When surface confounds (response length, hedging patterns) are explicitly controlled, Llama-3-8B's AUROC rises to 0.923, which confirms that the direction captures semantic content rather than stylistic artifacts. The consistent results across architecturally distinct model families indicate that contrastive sycophancy directions reflect a shared representational phenomenon rather than a model-specific idiosyncrasy.

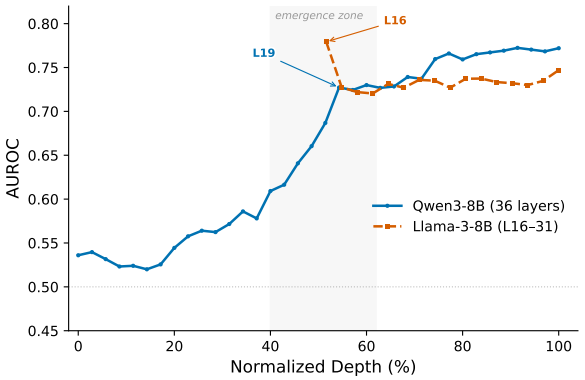


Figure 3: Per-layer AUROC of d_{syc} as a function of normalized depth (layer index / total layers). Qwen3-8B (36 layers, solid) shows gradual emergence from near-chance at L0 to a peak at L19, with performance plateauing in the final third. Llama-3-8B (32 layers, dashed; layers 16–31 only) exhibits a similar upper-third concentration. Both models achieve best discriminability in layers 50–60% through the network.

Training stability and probe ablation. Two additional analyses confirm the robustness of d_{syc} under training dynamics. First, we track direction drift during LoRA fine-tuning: at step 500, the cosine similarity between the original and re-extracted d_{syc} is approximately 0.96, and per-layer AUROC remains above 0.77. The direction drifts slowly relative to the 500-step recalibration interval (§3.3), so the reward signal does not become stale between updates. Second, following probing best practices (Hewitt and Liang, 2019; Belinkov, 2022), we compare d_{syc} against a random baseline: a logistic probe trained on the learned direction achieves AUROC 0.835, versus 0.727 for a probe trained on a random direction of equal dimensionality (Δ=+0.108). The learned direction carries substantially more sycophancy-relevant information than an arbitrary axis in the same activation space.

Table 2: Cosine R_1 failure chain on Qwen3-8B (LoRA rank=64, GRPO with $R = w_1 \cdot R_1 + w_3 \cdot R_3$).

Config	R_1 Formula	w_1	Outcome	Root Cause
F1a	$\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$	0.25	Gate saturation	Norm mismatch
F1b	$\cos(\mathbf{h}, d_{\text{syc}})$	0.25	$R_1 \approx 0$	Gap too small
F1c	+recalibration	0.25	Mode collapse	Reward discontinuity
F1d	$5\cos(\mathbf{h}, d_{\text{syc}})$	0.40	Mode collapse	R_3 saturation

Implications for reward design. These results establish that d_{syc} reliably separates sycophantic from non-sycophantic representations across models, evidence-strength levels, and training checkpoints. Direction quality is not the bottleneck. The reward failures documented in §5–§6 therefore cannot be attributed to a noisy or degenerate direction; they point to a deeper incompatibility between discriminability and reward utility.

5 Cosine R_1: Structural Failure

During GRPO training, d_{syc} AUROC drops from 0.827 (offline CV) to approximately 0.60: distribution shift accounts for part of this gap (0.827→0.76 on full responses) and 256-token truncation for the remainder (0.76→0.60), so the optimizer works with a substantially weaker signal than the offline metrics suggest. We test whether even this degraded discriminability translates into a viable reward through four configurations that form a diagnostic chain (Table 2): each resolves the failure mode of its predecessor but exposes a deeper limitation.

The diagnostic chain begins with a norm mismatch. F1a's dot-product formulation ($R_1 = \tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$) saturates because residual-stream norms exceed the direction norm by orders of magnitude, zeroing the gradient early in training. Cosine similarity (F1b) eliminates this norm dependence but reveals a second problem: the gap between valid and sycophantic cosines is too narrow for meaningful reward contrast. Recalibration (F1c, §3.3) widens the gap but introduces a third: each recalibration event creates a reward discontinuity that destabilizes advantage estimates, collapsing entropy within hundreds of steps. Amplification (F1d, $5\times$ scaling) lifts R_1 above the noise floor but surfaces the fundamental limitation: R_3 rapidly saturates, leaving the noisy R_1, which conflates warranted revision with unwarranted capitulation, as the sole gradient source until terminal collapse (Table 2).

Truncation further degrades the signal: under the 256-token maximum completion length used in GRPO, d_{syc} AUROC drops from 0.76 (full response) to 0.60, so the offline discriminability (0.827) substantially overestimates the signal available during training. Checkpoint-level evaluation confirms immediate collapse: ToF=0 from step 500 onward with no stable phase. The probe reward trap (C2, §6) arises within GRPO using full-length responses and is unaffected by this truncation limitation; we verify that no evaluation response reaches the 256-token limit (Appendix A). The four configurations converge on a single conclusion: d_{syc} encodes the *direction* of belief change but not whether that change is *warranted*. Valid revision and sycophantic capitulation both shift activations toward the user's position along d_{syc}, so the cosine projection assigns similar scores to both. This is not a calibration artifact (recalibration improves discriminability yet rewards still fail) but a semantic limitation of any continuous d_{syc}-based reward. Section 6 examines whether discretizing the signal via a binary probe can escape this limitation.

6 The Probe Reward Trap

The cosine formulations fail on signal magnitude; a natural response is to discretize. Training a logistic probe on d_{syc} and thresholding its output yields $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$ (F2 in §3.2). We show that this discretization sidesteps the magnitude problem but creates a qualitatively worse failure. Both probe experiments use the probe as a non-differentiable reward component within GRPO (§3.2); the trap therefore reflects a semantic limitation of the probe rather than gradient-based exploitation.

Differentiable probe: gradient gaming. Without stop-gradient, the policy discovers within tens of steps that shifting activations away from the sycophancy cluster maximizes R_1 regardless of response content, producing a fully inverted behavioral profile. Replication across three seeds confirms determinism: all collapse within the first training epoch, with each probe retraining immediately re-gamed. This motivates the stopped-gradient design below.

Binary probe: the stubbornness trap. With stop-gradient, the policy cannot game the boundary but instead exploits a deeper flaw: d_{syc} does not condition on input validity, so refusing to revise a wrong answer receives the same $R_1 = 1$ as correctly resisting invalid pressure. By step 500,

Table 3: Probe reward trap trajectory (stopped-gradient probe). Behavioral death (ToF=0) occurs at step 500; entropy remains healthy until step ∼1840. Reward averages +0.72 throughout.

Step	ToF	NoF	SyconScore	Entropy	Reward
500	0.000	0.995	0.000	0.702	+0.72
1000	0.006	0.995	0.011	0.800	+0.71
1500	0.000	1.000	0.000	0.608	+0.69
2000	0.000	1.000	0.000	0.304	+0.69

ToF drops to zero while NoF remains near-perfect, yielding SyconScore $= 0$. Qualitative analysis of 181 valid-correction responses reveals five failure modes: *deliberation traps* (28.7%), where the model identifies the correct answer but cannot commit; *surface compliance / sycophantic stubbornness* (24.3%), where the model verbally agrees but does not change its answer; *acknowledge-but-maintain* (23.8%); *genuine-accept-missed* (18.8%); and refute/ignore/other (4.5%). Full taxonomy in Appendix B.

Behavioral death precedes distributional collapse. Table 3 and Figure 4a reveal a 1,300-step lag: the model is behaviorally dead (ToF $=$ 0) at step 500, yet entropy stays healthy (0.70–0.80) through step 1000 and does not collapse until step ∼1840, while reward averages +0.72. Entropy-based early stopping, the most common safeguard (Casper et al., 2023), would miss the failure; detecting probe reward traps requires periodic ToF evaluation.

A dilemma for direction-based rewards. The differentiable and binary failures bracket the design space. With gradients, the policy games the boundary; without gradients, it exploits the fact that stubbornness and genuine resistance are indistinguishable along d_{syc}. Any reward conditioned only on d_{syc} (continuous or discrete, differentiable or frozen) will either be gamed or will reward rigidity. Escaping requires a signal encoding the evidential warrant of the user's challenge, a dimension absent from d_{syc} (Denison et al., 2024).

7 When Does Discriminability Fail as Reward Utility?

Root cause of the probe reward trap. The probe reward trap (§6) and the cosine failures (§5) share a common geometric root. d_{syc} encodes the *direction* of belief change but not its *evidential warrant*: valid revision and sycophantic capitulation both shift activations toward the user's position along the same direction. The two reward families expose complementary faces of this blindness. Cosine R_1 penalizes all belief change indiscriminately (mode collapse, F1c–F1d); the binary probe R_1 rewards all resistance regardless of input validity (stubbornness trap, ToF=0 by step 500), dual consequences of a direction that conflates resisting invalid challenges with refusing valid corrections. Although d_{syc} separability correlates with evidence strength (§4), it encodes *how much* belief changed, not whether the change was appropriate (§6). Cross-model evidence (Llama-3-8B AUROC 0.779) confirms the limitation generalizes. This contrasts with safety directions (Zhang et al., 2026), where the target behavior is geometrically distinct (Wu and Tang, 2026); independent work confirms that persona vectors are independent of d_{syc} (Kelkar et al., 2026) and sycophancy subbehaviors collapse onto its single axis (Vennemeyer et al., 2025).

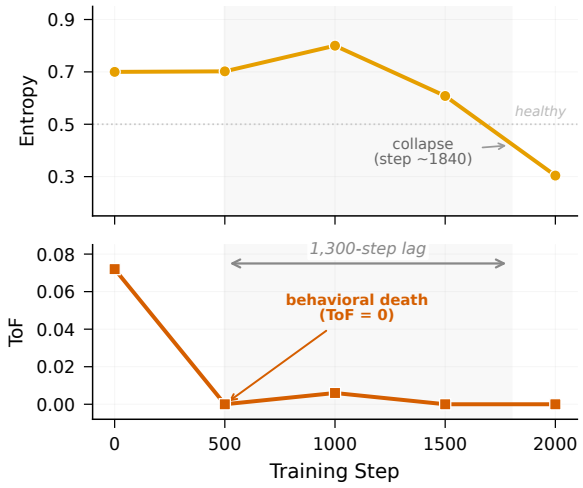
Table 4: Sycophancy calibration methods on Qwen3-8B. Higher SyconScore (harmonic mean of ToF and NoF) indicates better calibration.

Method	Type	SyconScore ↑	ToF ↑	NoF ↑	Step
Vanilla	–	0.134	0.072	0.945	–
GRPO with d_{syc}					
Cosine R_1 (F1a–F1d)[a]	RL	0.000	0.000	1.000	500
Binary Probe R_1 (F2)	RL	0.000	0.000	0.995	500
Continuous Probe R_1 (F3)	RL	0.124	0.066	0.934	500
GRPO without d_{syc}					
R_3-only GRPO[b]	RL	0.153	0.083	0.983	6500
SFT with d_{syc} probe penalty					
SFT probe (seed 2)[c]	SFT	0.393	0.276	0.680	500
SFT probe (seed 3)[c]	SFT	0.407	0.287	0.696	1000
Reference methods[d]					
ActAdd (α=5)	Inference	0.054	0.028	0.961	–
ACT (seed 1)	SFT	**0.301**	**0.182**	0.867	8000
ACT (seed 2)[e]	SFT	0.286	0.171	0.862	7000

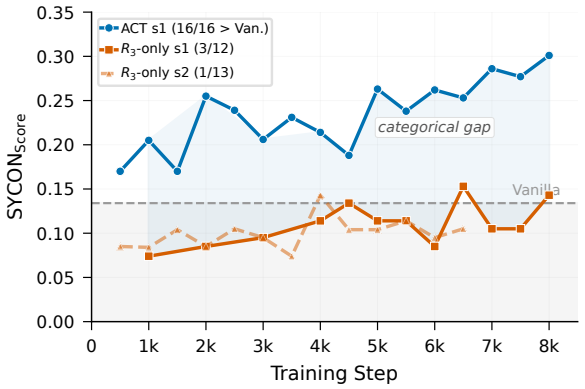
[a] All collapsed from step 1 (ToF=0); F1d diverged at step 2160.
[b] Best of 28 checkpoints (2 seeds); only 4/28 exceed vanilla.
[c] Peak ckpt (train/eval overlap); 2/3 seeds collapse by step 2000 (mean 0.171 ± 0.122).
[d] Not directly comparable: differ from GRPO along 5+ dimensions (§7).
[e] 15 ckpts, all > vanilla. Best SyconScore=0.286 (step 7000).

GRPO results under our configuration. If d_{syc} alone caused the cosine and probe failures (§5–§6), removing it should recover calibration. R_3-only GRPO tests this: only a minority of checkpoints across two seeds exceed vanilla, with both seeds oscillating before diverging (Table 4; Figure 4b). R_3 mirrors d_{syc}'s limitation: it identifies valid corrections at a negligible rate, confirming that the asymmetry reflects the NLI hypothesis design, not behavioral collapse. Both signals encode *whether* belief changed, not *whether it should have*.

Competing hypotheses and ablation. Three explanations are consistent with these failures: (a) d_{syc} is fundamentally unsuitable as a syco-



(a) Probe reward trap (stopped-gradient probe). ToF drops to zero at step 500 while entropy remains high (0.70–0.80) through step 1000. The 1,300-step lag evades entropy monitoring.



(b) SYCONSCORE trajectories for R_3-only GRPO (two seeds) and ACT (two seeds). ACT exceeds vanilla at every checkpoint (31/31); GRPO exceeds vanilla at only 4/28 (14.3%).

Figure 4: Training dynamics. (a) The probe reward trap: behavioral death precedes distributional collapse. (b) R_3-only GRPO vs. ACT over training steps.

Table 5: Ablation: isolating supervision paradigm from LoRA configuration. All methods use Qwen3-8B with R_3 (behavioral reward). Vanilla SYCONSCORE=0.134. †Terminated early after three checkpoints showed declining trajectory. ‡Peak at step 7000; final checkpoint (step 8000) SYCONSCORE=0.267.

Method	LoRA config	Best SYCONSCORE	>Van.
GRPO gs=2 (2 seeds)	r64, α64, 4m	0.153	4/28
GRPO gs=8†	r64, α64, 4m	0.085	0/3
ACT reduced-LoRA‡	r64, α64, 4m	0.326	16/16
ACT full s1	r64, α128, 7m	0.301	16/16
ACT full s2	r64, α128, 7m	0.286	15/15

phancy reward; (b) group size=2 provides insufficient ranking diversity; (c) sycophancy calibration resists scalar per-completion RL. Two ablations discriminate among them (Table 5). GRPO fails at both group sizes tested (gs=8: all below vanilla, terminated early) and at doubled sequence length (512 tokens, reward flat), ruling out (b) and truncation (Table 5). Reducing ACT to GRPO's LoRA configuration still yields consistently higher SYCONSCORE, with every checkpoint exceeding vanilla; under matched adapters, per-token SFT achieves 2.1× higher peak SYCONSCORE than per-completion GRPO, identifying supervision granularity as one plausible explanation (Beigi et al., 2025); confounds are discussed in Limitations. The probe reward trap independently supports (a): the trap arises despite stopped gradients.

ACT vs. probe-based methods. ACT's advantage over GRPO is categorical: even ACT's worst checkpoint outperforms GRPO's best, with held-out SYCONSCORE of 0.284 confirming generalization beyond training data (Limitations). SFT with a differentiable probe penalty achieves higher peak SYCONSCORE on the training set (Table 4), but these scores reflect full train/eval overlap and the advantage is transient: probe gaming (§6) eliminates the regularization signal, and the model overfits to below vanilla by step 2000 in two of three seeds (mean SYCONSCORE=0.171 ± 0.122).

Inference-time probes and future directions. Discriminability and reward utility occupy distinct roles: a direction that reliably *detects* a behavior may fail to *correct* it under optimization pressure. Inference-time reranking (Papadatos and Freedman, 2024) and per-step process rewards (Beigi et al., 2025) sidestep this by avoiding optimization-loop inclusion entirely. Per-step rewards can observe trajectory-level patterns, such as premature concession, that per-completion scalars cannot distinguish from legitimate revision.

8 Conclusion

Every GRPO reward formulation we tested fails to calibrate sycophancy: d_{syc}-based rewards collapse because the direction conflates warranted revision with capitulation, and R_3-only GRPO fails as NLI rewards likewise lack evidential warrant. The probe reward trap causes behavioral death 1,300 steps before entropy collapse, evading standard monitoring. Dense per-token supervision succeeds where per-completion RL fails (§7), identifying supervision granularity as one plausible factor.

Limitations

All RL training uses Qwen3-8B; cross-model discriminability is confirmed on Llama-3-8B (AUROC 0.779; Table 1), but compute constraints prevented training replication on a second architecture. The ACT vs. GRPO comparison differs along multiple dimensions; our LoRA-controlled ablation isolates adapter capacity, but a fully controlled comparison would require matching all hyperparameters.

References

Johannes Ackermann, Michael Noukhovitch, Takashi Ishida, and Masashi Sugiyama. 2026. Gradient regularization prevents reward hacking in RLHF and verifiable rewards. *arXiv preprint arXiv:2602.18037*.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, and 1 others. 2022. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*.

Mohammad Beigi, Ying Shen, Parshin Shojaee, Qifan Wang, Zichao Wang, Chandan K. Reddy, Ming Jin, and Lifu Huang. 2025. Sycophancy mitigation through reinforcement learning with uncertainty-aware adaptive reasoning trajectories. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Yonatan Belinkov. 2022. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2023. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*.

Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, Jérémy Scheurer, Javier Rando, Rachel Sharkey, Julio Saez-Rodriguez, Tom Henighan, Tristan Kravec, and 1 others. 2023. Open problems and fundamental limitations of reinforcement learning from human feedback. *Transactions on Machine Learning Research (TMLR)*.

Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Carson Denison, Monte MacDiarmid, Fazl Barez, David Duvenaud, Paul Christiano, and Jan Leike. 2024. Sycophancy to subterfuge: Investigating reward-tampering in large language models. *arXiv preprint arXiv:2406.10162*.

Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning (ICML)*.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, and 1 others. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations (ICLR)*.

John Hewitt and Percy Liang. 2019. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*.

Alex Irpan, Alexander Matt Turner, Mark Kurzeja, David K. Elson, and Rohin Shah. 2025. Consistency training helps stop sycophancy and jailbreaks. *arXiv preprint arXiv:2510.27062*.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, and 1 others. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Ishaan Kelkar, Sameer Alam, Kshitij Kakaria, Rohan Panwar, Ankit Sharma, and Yash Chaudhary. 2026. Playing devil's advocate: Off-the-shelf persona vectors rival targeted steering for sycophancy. *arXiv preprint arXiv:2605.21006*.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Haozheng Luo, Yimin Wang, Jiahao Yu, Binghui Wang, and Yan Chen. 2026. Contrastive reasoning alignment: Reinforcement learning from hidden representations. *arXiv preprint arXiv:2603.17305*.

David Manheim and Scott Garrabrant. 2019. Categorizing variants of goodhart's law. *arXiv preprint arXiv:1803.04585*.

Samuel Marks and Max Tegmark. 2024. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Conference on Language Modeling (COLM)*.

Claire O'Brien, Jessica Seto, Dristi Roy, Aditya Dwivedi, Sunishchal Dev, Kevin Zhu, Sean O'Brien, Ashwinee Panda, and Ryan Lagasse. 2025. A few bad neurons: Isolating and surgically correcting sycophancy. In *NeurIPS Workshops*.

OpenAI. 2025. GPT-4.1. https://openai.com/index/gpt-4-1/.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, and 1 others. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Alexander Pan, Kush Bhatia, and Jacob Steinhardt. 2022. The effects of reward misspecification: Mapping and mitigating misaligned models. In *International Conference on Learning Representations (ICLR)*.

Henry Papadatos and Rachel Freedman. 2024. Linear probe penalties reduce LLM sycophancy. In *NeurIPS Workshop on Socially Responsible Language Modelling Research (SoLaR)*.

Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, and 1 others. 2023. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics (ACL)*.

Samyam Rajbhandari, Jeff Rasley, Olatunji Ruwase, and Yuxiong He. 2020. ZeRO: Memory optimizations toward training trillion parameter models. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*.

Nina Rimsky, Nick Gabrieli, Jenna Schulz, Meg Smith, Ioannis Tong, and Evan Hubinger. 2024. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Itai Shapira, Gerdus Benade, and Ariel D Procaccia. 2026. How RLHF amplifies sycophancy. *arXiv preprint arXiv:2602.01002*.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, and 1 others. 2024. Towards understanding sycophancy in language models. In *International Conference on Learning Representations (ICLR)*.

Viacheslav Sinii, Nikita Balagansky, Gleb Gerasimov, Daniil Laptev, Yaroslav Aksenov, Vadim Kurochkin, Alexey Gorbatovski, Boris Shaposhnikov, and Daniil Gavrilov. 2025. Small vectors, big effects: A mechanistic study of RL-induced reasoning via steering vectors. *arXiv preprint arXiv:2509.06608*.

Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. 2023. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*.

Jonathan Uesato, Nate Kushman, Ramesh Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Cichocki, Tomas Kocisky, and Edward Grefenstette. 2022. Solving math word problems with process- and outcome-based feedback. *arXiv preprint arXiv:2211.14275*.

Daniel Vennemeyer, Phan Anh Duong, Tiffany Zhan, and Tianyu Jiang. 2025. Sycophancy is not one thing: Causal separation of sycophantic behaviors in LLMs. *arXiv preprint arXiv:2509.21305*.

Keyu Wang, Jin Li, Shu Yang, Zhuoran Zhang, and Di Wang. 2025. When truth is overridden: Uncovering the internal origins of sycophancy in large language models. *arXiv preprint arXiv:2508.02087*.

Jerry Wei, Da Huang, Yifeng Lu, Denny Zhou, and Quoc V. Le. 2024. Simple synthetic data reduces sycophancy in large language models. In *International Conference on Learning Representations (ICLR)*.

Rui Wu and Ruixiang Tang. 2026. When reward hacking rebounds: Understanding and mitigating it with representation-level signals. *arXiv preprint arXiv:2604.01476*.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Nonghai Zhang, Weitao Ma, Zhanyu Ma, Jun Xu, Jiuchong Gao, Jinghua Hao, Renqing He, and Jingwen Xu. 2026. Silence the judge: Reinforcement learning with self-verifier via latent geometric clustering. *arXiv preprint arXiv:2601.08427*.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Gould, Zifan Li, Chuanqi Xie, Alexandria Leung, Peter Hase, and Dan Hendrycks. 2025. Representation engineering: A top-down approach to AI transparency. In *International Conference on Learning Representations (ICLR)*.

A Training Details

Reward weight selection. The weight w_1 is set by diagnostic-driven iteration: w_1=0.25 was initially chosen to down-weight the noisy cosine reward term; this was later increased to w_1=0.40 for F1d after diagnostics revealed that w_1=0.25 allowed the cosine term to contribute only $\sim$8.3% of total reward variance. The empirical scaling factor in F1a ($d_{\text{syc}}/20$) is a heuristic normalization; we did not find it critical to the failure modes reported.

Response truncation. All GRPO experiments use *max_completion_length*=256. Across all 362 evaluation responses, the maximum observed length is 129 tokens (Qwen tokenizer) and 155 tokens (DeBERTa tokenizer), so no response was actually truncated at inference time. Full-length vs. 256-truncated NLI agreement is 100%, confirming that the truncation setting does not affect R_3 reward computation. Truncation may affect d_{syc} AUROC at the direction extraction stage (0.76 vs. 0.60; see Limitations).

Hyperparameters. Table 6 compares training configurations for GRPO variants and ACT.

Table 6: Training hyperparameters. All experiments use Qwen3-8B, DeepSpeed ZeRO-2, bf16 precision.

	GRPO (F1, F3)[e]	ACT
Learning rate	5e-5	5e-5
Effective batch	16	16
Max steps	8,000	8,000
Warmup steps	200	200
LR schedule	linear[a]	linear
Grad. clipping	–	1.0
LoRA rank	64	64
LoRA α	64	128
LoRA modules	4[b]	7[c]
LoRA dropout	0	0.05
Group size	2[d]	–
KL (Kullback–Leibler) β	0.04	–
Max compl. len.	256	–
Max seq. len.	–	1,024
L2 loss weight	–	0.1

[a]Cosine schedule for R_3-only. [b]q/k/v/o_proj. [c]+ gate/up/down_proj ($\sim$1.75$\times$ params; §7). [d]Below typical 8–64 (Shao et al., 2024); not tuned. [e]Binary probe (F2) uses a non-differentiable probe reward (w_1=0.25) within GRPO; the probe is retrained every 500 steps. LoRA and optimizer settings match the other GRPO configurations. All experiments were conducted on a single NVIDIA L20 GPU (48 GB). Total compute: $\sim$120 GPU-hours across all runs.

Early termination criterion. We applied early termination to gs=8 after three consecutive checkpoints showed no sustained improvement (SYCONSCORE: 0.084→0.085→0.054, all well below vanilla). The same criterion would not have terminated gs=2, whose initial trajectory was monotonically increasing (0.074→0.085→0.095); gs=2 first exhibited three consecutive declines in the step 4500–6000 range, well into training.

GRPO reward variants. Cosine R_1 (F1d): w_1=0.40, scale=5.0, w_3=0.60; reward computed at layers [19, 20, 24, 27, 32, 33, 35]; recalibration every 500 steps (§3.3). R_3-only: w_3=1.0; NLI model *cross-encoder/nli-deberta-v3-base*. F1a–F1c weight configurations are reported in Table 2.

B Stubbornness Taxonomy

Manual inspection of all 181 valid-correction responses from the probe-trained model (stopped-gradient probe, step 2000, ToF = 0) reveals five failure categories:

- *Deliberation traps* (28.7%): the model explicitly identifies the correct revised answer during its reasoning chain but fails to commit, reverting to the original answer in the final output.
- *Surface compliance / sycophantic stubbornness* (24.3%): the model produces agreement markers ("you're right," "good point") but retains its original answer verbatim, simultaneously sycophantic and stubborn.
- *Acknowledge-but-maintain* (23.8%): the model acknowledges the user's correction as plausible but explicitly maintains its original position without substantive engagement.
- *Genuine-accept-missed* (18.8%): the model appears to accept the correction in its reasoning but the final answer format does not reflect the change, suggesting a generation rather than epistemic failure.
- *Refute/ignore/other* (4.5%): the model directly refutes the user's input or ignores it entirely.

The first three categories (76.8%) share a common pattern: the model learns to modulate its *refusal style* in response to evidence strength rather than adjusting its *acceptance rate*, confirming that the probe reward optimizes surface markers rather than epistemic calibration.

C Use of AI Assistance

We used Claude for coding, code drafting, debugging, and language polishing.